Speech Emotion Recognition from Acoustic Features: A Speaker-Independent Study of Noise Robustness

Corinne D'Elia
Department of Computer Science
Università degli Studi di Milano
Milan, Italy

***Abstract*—This study investigates the classification of four acted emotions from speech and the effect of background noise on recognition. Calm, happy, sad, and angry recordings from the RAVDESS dataset are represented by 101 acoustic features. The pipeline includes Principal Component Analysis (PCA), k-means clustering, and a comparison of Support Vector Machine (SVM) and Random Forest (RF) classifiers. All training and test actors are separate. On clean speech, SVM obtains a macro-F1 of 0.731 and RF obtains 0.708; the difference is uncertain when speakers are resampled. The trained models are also tested with white, traffic, and cafeteria noise at 20, 10, and 0 dB. RF achieves a higher macro-F1 in all nine noisy conditions, although both models lose accuracy. Permutation importance shows that MFCCs are the most influential feature group. The results highlight the need to test both new speakers and noisy environments.**

***Index Terms*—speech emotion recognition, RAVDESS, acoustic features, support vector machine, random forest, noise robustness, explainable AI**

I. Introduction

Speech Emotion Recognition (SER) aims to infer emotional expression from vocal signals. A speaker can change pitch, intensity, rhythm, and the distribution of spectral energy without changing the words being spoken. Recognizing these differences can support the study of spoken interaction and the development of speech-based interfaces. However, a classifier may also learn characteristics of individual speakers or recording conditions instead of information that is useful for new voices.

This project asks two questions: *How well can acoustic features distinguish four emotions in voices not heard during training?* and *How much does this performance change when background noise is added?* To answer them, we compare an SVM and a Random Forest using the same speaker-independent test set. We also examine the structure of the feature space with PCA and k-means and use permutation importance to help interpret the predictions.

The evaluation is deliberately focused on traditional machine learning rather than deep neural networks. This choice allows a direct comparison of two different classifiers using interpretable acoustic descriptors. The main additional experiment is a controlled noise test: after training only on clean speech, both classifiers receive noisy versions of the same test recordings. The resulting change in macro-F1 indicates how sensitive the models are to conditions not present during training. The analysis follows the audio feature extraction, clustering, classification, and explainability steps of the Audio Pattern Recognition course.

II. Data and Experimental Protocol

A. Speech and noise recordings

The speech data are taken from the *audio-only speech* part of the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS). It contains acted emotional expressions spoken by 24 actors. We select four labels: *calm*, *happy*, *sad*, and *angry*. The original selection contains 768 candidate recordings, with 192 for each class. An audio-content hash check finds one duplicate pair within the same actor–emotion combination. After keeping one copy, the experiment uses 767 distinct recordings: 192 calm, 191 happy, 192 sad, and 192 angry.

For the noise experiment, we use synthetic white Gaussian noise and two real environmental recordings from the DEMAND database: *STRAFFIC* (traffic) and *PCAFETER* (cafeteria). The environmental noises are taken from the `ch01.wav` channels of the 16-kHz recordings. These three conditions represent different types of interference, but they do not cover every possible real acoustic environment.

B. Independent speakers and validation

Six actors (IDs 1, 2, 15, 17, 18, and 24; three men and three women) are reserved for the final test. The other 18 actors are used for model development. This produces 575 development recordings and 192 test recordings, with no actor present in both groups. The test set has 48 examples per emotion.

Model and feature choices use three-fold `GroupKFold` cross-validation on the development data. Actor identity is the group, so validation actors never occur in the corresponding training fold. Missing-value imputation, feature standardization, and optional PCA are placed inside each learning pipeline, which means their parameters are learned only from the training portion of a fold. The separate test set is not used to tune the algorithms.

III. Audio Processing and Feature Extraction

A. Signal representation

Audio recordings are converted to mono, resampled to 16 kHz if required, and centered to remove the DC offset. Each waveform is analyzed by the Short-Time Fourier Transform (STFT), with a Hann window of 512 samples (32 ms) and a

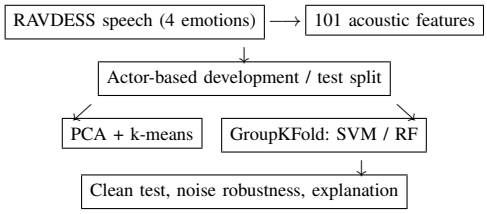


Fig. 1. Overview of the experimental workflow. All supervised model choices are made without using the final test actors.

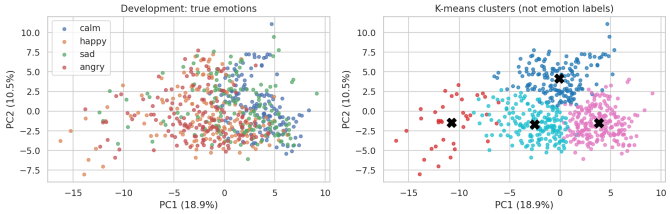


Fig. 2. Two-dimensional PCA view of the development recordings: true emotions (left) and k-means clusters (right). The labels overlap in feature space.

hop of 160 samples (10 ms). A 64-band Mel filter bank is applied to the power spectrum; taking its logarithm produces a Log-Mel spectrogram. The Mel representation is then used to obtain 13 Mel-Frequency Cepstral Coefficients (MFCCs). These descriptors summarize important characteristics of the vocal spectrum.

The complete representation contains 101 features per recording. There are 78 MFCC-related descriptors: means and standard deviations of 13 static MFCCs, 13 first derivatives (delta), and 13 second derivatives (delta-delta). A further 16 descriptors measure energy, zero-crossing rate, and spectral properties, including centroid, bandwidth, rolloff, flatness, entropy, and flux. The remaining seven describe pitch and periodicity. Pitch is estimated using autocorrelation, subject to basic validity checks.

For time-varying quantities, we mainly compute means and standard deviations over frames. This creates one fixed-length vector for each audio recording, suitable for SVM and RF. The main limitation is that temporal order and local changes are only partly preserved. Silences are not removed and amplitude is not normalized; energy and pauses can therefore carry information but may also be affected by added noise. Waveforms and Log-Mel spectrograms were inspected for examples of each emotion produced by the same actor with matching recording settings.

B. Feature space exploration

To describe the unsupervised structure of the data, we apply imputation, standardization, and PCA to the *development* features. PCA maps correlated descriptors into orthogonal directions of decreasing variance. Keeping at least 95% of total variance requires 52 components in this experiment. The first two components provide a two-dimensional view, but the clustering is run in the full retained PCA space.

We then apply k-means with $k = 4$, equal to the number of studied emotions, without using emotion labels to form the clusters. A silhouette score of 0.107 indicates limited separation. The Adjusted Rand Index (ARI) is 0.105 and Normalized Mutual Information (NMI) is 0.133, showing weak agreement between the clusters and the four emotion labels. As shown in Fig. 2, the emotions overlap substantially in the first two components. This does not rule out stronger supervised classification, because supervised models use the actual class labels.

IV. Supervised Classification

A. Model selection

The first classifier is an SVM, with both linear and radial basis function (RBF) kernels examined during tuning. The second is a Random Forest, which combines multiple decision trees. Both are tested with (i) 26 mean/standard-deviation descriptors from the 13 static MFCCs and (ii) the complete 101-feature representation. Grid search selects hyperparameters by mean *macro-F1* over three speaker-based folds. Macro-F1 is the mean of class-specific F1 scores, so each emotion has the same importance in evaluation even if the dataset contains small class-count differences.

The full feature set performs better for both models: development cross-validation macro-F1 is 0.610 for SVM and 0.605 for RF, compared with 0.517 and 0.504 with MFCC-only features. The selected SVM has an RBF kernel, $C = 10$, and no PCA step; the RF uses 200 trees, `max_features=sqrt`, and `min_samples_leaf=3`. The chosen models are refitted on the full development set before testing.

There is a large training–validation gap. Mean training macro-F1 is 1.000 for SVM and 0.996 for RF, while validation scores are 0.610 and 0.605. This suggests substantial overfitting or difficulty transferring to unseen voices. The speaker-independent protocol makes this problem visible rather than hiding it with a random clip-level split.

B. Performance on clean speech

Table I reports results on the 192 untouched test recordings. Both models improve clearly on a majority-class Dummy baseline. The SVM scores 73.44% accuracy and 0.731 macro-F1; RF scores 72.40% accuracy and 0.708 macro-F1. Because the test includes only six actors, the paired difference is also studied by actor-block bootstrap: entire groups of recordings are resampled together, rather than treating clips from one person as independent observations. The SVM minus RF macro-F1 difference is 0.023 with a descriptive 95% interval of $[-0.097, 0.129]$. Since this interval includes zero, the clean test does not provide strong evidence that one method is consistently superior.

The class-level errors are not uniform (Fig. 3). Recall for *angry* is about 85% for SVM and 88% for RF. For *sad*, recall is only about 54% and 38%, respectively. This shows why macro-F1 and confusion matrices are more informative than accuracy alone.

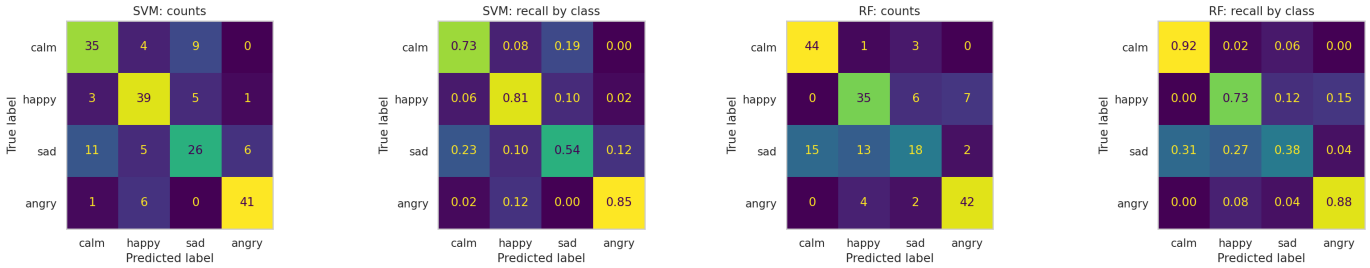


Fig. 3. Clean-test confusion matrices for SVM (left) and Random Forest (right). Each original figure contains counts and per-class recall. The models identify *angry* better than *sad*.

TABLE I
RESULTS ON THE CLEAN, SPEAKER-INDEPENDENT TEST SET (192 CLIPS).

Model	Accuracy	UAR	Macro-F1
SVM (RBF)	0.734	0.734	0.731
Random Forest	0.724	0.724	0.708
Dummy baseline	0.250	0.250	0.100

TABLE II
MEAN MACRO-F1 ACROSS THREE NOISE REALIZATIONS FOR EACH SNR.

Noise	Model	20 dB	10 dB	0 dB
White	SVM	0.176	0.100	0.100
	RF	0.517	0.398	0.277
Traffic	SVM	0.303	0.184	0.103
	RF	0.596	0.385	0.189
Cafeteria	SVM	0.318	0.219	0.137
	RF	0.638	0.477	0.313

V. ROBUSTNESS TO BACKGROUND NOISE

A. Construction of noisy audio

The trained models are tested without retraining or noise augmentation. For a clean signal s and a zero-mean noise segment n, a mixture is generated as

$$x = s + \alpha n, \qquad \alpha = \sqrt{\frac{P_s}{P_n \, 10^{\mathrm{SNR}/10}}}, \tag{1}$$

where P_s and P_n are average powers over the entire clip. The chosen Signal-to-Noise Ratios are 20, 10, and 0 dB; lower values indicate stronger interference. At 0 dB, speech and noise have equal average power. The same noise segment is reused across SNR levels within a repetition, changing only its amplitude. The analysis verifies the achieved SNR numerically.

For each of 192 test clips, the experiment generates three noise types, three SNR levels, and three independent noise realizations: $192 \times 3 \times 3 \times 3 = 5{,}184$ mixtures. Both models classify every mixture, producing 10,368 predictions. Mean macro-F1 and standard deviation are calculated across the three repetitions. The measured SNR includes silent parts of the recordings and should not be interpreted as the SNR within active speech only.

B. Results under noise

The performance curves in Fig. 4 show a consistent pattern: RF has higher mean macro-F1 than SVM for every noise type at all three SNRs. At cafeteria noise and 20 dB, RF reaches 0.638 while SVM reaches 0.318. At 10 dB traffic noise, the results are 0.385 and 0.184. With 0 dB cafeteria noise, the scores fall to 0.313 (RF) and 0.137 (SVM). Table II provides the complete comparison. The drop from clean conditions is not the same across models, despite their similar clean-test results.

A particularly clear failure appears for SVM with white noise at 0 dB. In the first realization, it predicts *sad* for all 192 recordings. This yields 25% accuracy but only 0.100 macro-F1, as three emotions have zero recall. The same macro-F1 value is observed for all three repetitions at 10 and 0 dB, although the predicted class should not be inferred from the score alone. The 0-dB confusion matrices for the environmental noises also show a strong concentration of SVM predictions toward *sad*. This is compatible with a shift in acoustic feature distributions between clean training audio and noisy test audio, but it does not prove which features cause the failures.

VI. EXPLAINABILITY AND DISCUSSION

A. Permutation feature importance

Permutation importance measures the decrease in test macro-F1 when selected features are shuffled across recordings while the trained model remains fixed. The experiment first tests individual descriptors, then shuffles complete acoustic families using a common permutation to keep within-family relationships. The groups are MFCCs, delta, delta-delta, energy/zero-crossing rate, pitch/periodicity, and spectral descriptors.

MFCCs are the most influential group for both classifiers. Their permutation decreases macro-F1 by about 0.290 for SVM and 0.133 for RF. The next strongest group is delta for SVM (0.125) and energy/zero-crossing rate for RF (0.071). These values describe reliance on features in the *clean test set*; they are not evidence of a causal relationship with emotion and do not directly explain behavior under noise. Groups also have different numbers of descriptors, so group importance is not importance per individual feature.

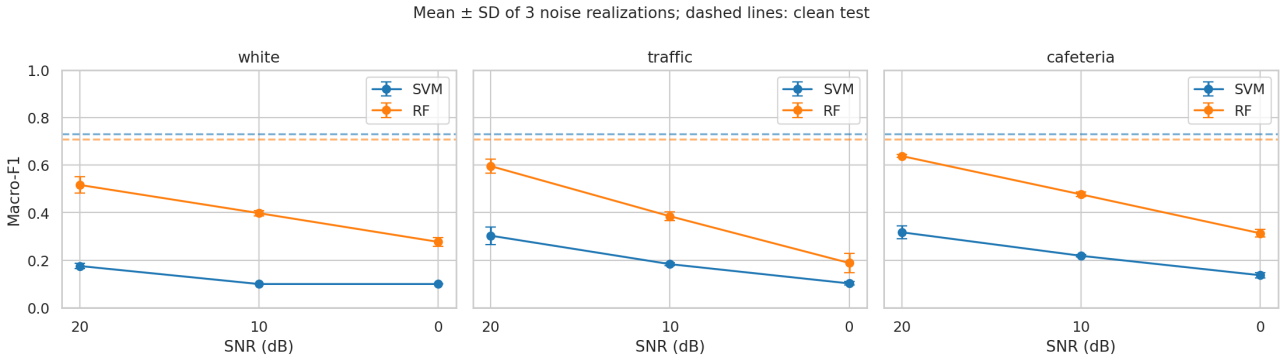


Fig. 4. Mean macro-F1 for the three noise types and three SNR levels (error bars: standard deviation over three noise realizations). Dotted lines show each model's clean-test macro-F1.

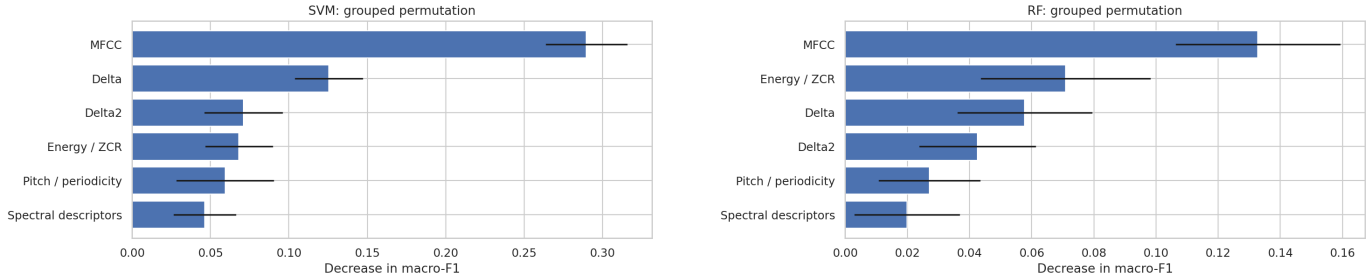


Fig. 5. Decrease in clean-test macro-F1 after shuffling each feature family: SVM (left) and RF (right). Error bars show variation over repeated permutations.

B. Interpretation and limitations

The combined results suggest that richer acoustic features are useful for recognizing acted emotions from voices not seen during model development. The weak k-means alignment indicates that the four emotional categories are not naturally separated into compact groups, even though supervised methods reach much higher classification scores. The small and uncertain clean-test difference between SVM and RF is less important than their contrasting behavior after noise is added: RF is more robust in this specific experimental protocol.

Several limits are important. First, RAVDESS contains acted speech in controlled conditions, not spontaneous conversation; results should not be directly generalized to natural emotional speech. Second, only six test actors are available, limiting confidence about other speakers. Third, models are trained on clean recordings only: neither denoising nor noise-augmented training is examined. Fourth, averaging acoustic features over time removes some prosodic dynamics. Finally, the experiment uses three noise types, three SNR levels, and three realizations, so its results do not cover all real recording conditions. These limits support cautious interpretation of the reported numbers.

VII. Conclusion

This work presents a speaker-independent speech emotion recognition pipeline using acoustic feature extraction, PCA, k-means, SVM, RF, and permutation importance. Both classifiers perform substantially better than a simple baseline on clean recordings. However, their results change strongly when noise is introduced. The Random Forest shows consistently better performance than SVM across all nine noisy conditions, despite a slightly lower clean-test score. The central finding is therefore not only which model performs best on clean data, but also how much its performance depends on acoustic conditions. Future experiments could test noise-aware training and additional speakers while keeping the same speaker-independent evaluation protocol.